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PROJECT REPORT - AERODYNAMICS OF TRANSPORT VEHICLES - VEHICLES AERODYNAMICS

Aerodynamic devices on a truck without trailer

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1. Introduction

The aerodynamic efficiency plays a crucial role in the design of transport vehicles, particularly when discussing heavy-duty trucks. Since these vehicles operate over long distances and the aerodynamic drag accounts for more than 50% of the required power, its reduction has a substantial impact on fuel consumption and, as a consequence, on the overall operating costs of this transportation method. For this reason, several devices are adopted to improve their aerodynamic performance, especially by acting on the gap between the cabin and the trailer, which represents one of the main sources of drag for trucks [1]. Among them, the most effective solution consists in the use of side and roof fairings, as shown in Figure 1.

Although the tractor is connected to the trailer for most of its operating time, there are instances in which it moves without it, even if only for short distances. Thus, it can be of interest to analyze the effect of such devices, designed to improve the performance of the full vehicle, on the cabin whenever the trailer is removed.

The study was conducted through CFD simulations, considering multiple configurations. First, the drag produced by the cabin was evaluated with and without the fairings to obtain a preliminary assessment. Then, each device was analyzed individually to determine its specific con-

tribution to the aerodynamic behavior. Once the fundamental flow physics was understood, different configurations were considered in order to reduce the overall drag, ultimately leading to a final solution that significantly improves the aerodynamic resistance.

2. Vehicle Geometry

The CAD model employed for the analysis was created by taking as a reference the *EcoJeT-truck* geometry¹, a common European-type truck. Since this model has a great amount of detail, it was considerably simplified using the 3D modeling software *CATIA V5* from *Dassault Systèmes* without losing the main features that are meaningful for the analysis. This is particularly important to make it suitable for CFD analysis, given the limited computational resources available.

To be specific, the entire cooling system and the brackets that hold the mirrors were removed, as well as the blind spot mirrors. Furthermore, the underbody was simplified, creating a completely flat surface, since this region does not significantly affect the case of interest. The back of the cabin was simplified in the same fashion. Additionally, all small cabin details were not considered and the thickness of the fairings was in-

¹Provided on the *WeBeep* page of the course.

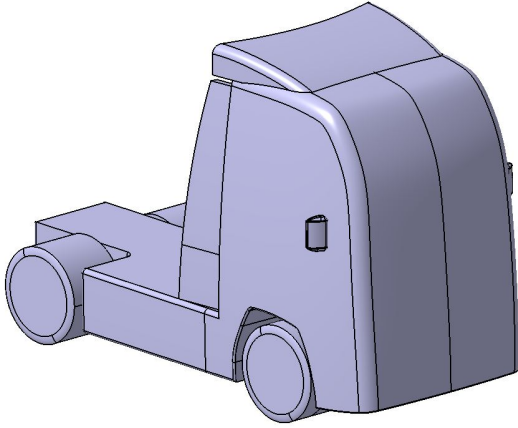


Figure 1: Truck CAD model

creased to ease the realization of the mesh and the convergence of the simulations. Another important feature is the choice of the ratio between the radius of the cabin front corners and the cabin spanwise width $\frac{r}{b} = 0.075$, which allows for a reduction in flow separation at the corner. Finally, the wheels were made completely flat on the side [2].

The main dimensions of the truck model are reported in Table 1.

Length [m]	Width [m]	Height [m]	Frontal Area [m ²]
6	2.4	4	4.52

Table 1: Truck dimensions

3. Computational analysis

The numerical simulations were performed using the open source CFD software *OpenFOAM*.

In all cases, a free stream velocity $U_\infty = 20 \text{ m/s}$ was set, which corresponds to a Mach number of approximately 0.06. This means that the problem can be considered incompressible with reasonable approximation, since the Mach number is much lower than 0.3. The Reynolds number is about 6.6×10^6 . Notice that the chosen value of the velocity is quite representative of actual road speeds for heavy-duty trucks.

3.1. Numerical set-up

In order to reduce the computational cost, all simulations were performed by considering only half of the full domain, exploiting the symmetry

of the geometry with respect to the longitudinal plane. This choice is coherent with the aim of the analysis, since the mean forces applied to the vehicle are not sensitive to the imposed symmetry constraint.

The problem considered involves a bluff body at high Reynolds number, characterized by large separation regions, which would require an unsteady solver to capture the behavior of the flow; however, given the limited computational power available, it was decided to model the problem with steady-state incompressible *RANS* [3] equations, which are common when dealing with external aerodynamics problems.

The *RANS* equations were solved by means of the incompressible solver *simpleFoam* and the $k-\omega$ *SST* turbulence model with wall functions². This model allows for a good representation of the turbulent flow, both close to and far from the wall. In order to obtain values of y^+ within the interval between 30 and 300, four cell layers were added to both the truck and the lower wall.

The simulations were initiated with the *potentialFoam* solver in order to obtain better initial conditions for *simpleFoam*, leading to faster convergence. The simulation convergence criterion was configured with a maximum number of 1000 iterations; in fact, it was observed that beyond this point, the drag coefficient value stabilized, while the main quantities residual reached a 10^{-3} order of magnitude.

In Table 2, the boundary conditions for all main quantities on each patch are reported.

Notice that, in order to limit the computational cost, wheels rotation was neglected, since it does not affect the regions of interest for the analysis; therefore, it can be expected that relatively low drag coefficient values will be obtained compared to experimental results. Conversely, the ground speed was set with a relative velocity with respect to the model to match U_∞ .

3.2. Mesh

The external domain was created using the *blockMesh* application from *OpenFOAM*, which allows for its discretization by means of cubic cells with an imposed edge length.

Then, the final mesh was obtained by employing

²Average $y^+ \approx 70$ over the whole truck and $y^+ \approx 200$ on the lower wall.

Patch	U	P	k	ω
Tractor	Stationary Wall no-slip	zeroGradient	kqRWallFunction	omegaWallFunction
Inlet	Uniform (20 0 0)	zeroGradient	Uniform	Uniform
Outlet	inletOutlet	Pressure Outlet	inletOutlet	inletOutlet
Upper Wall	Stationary Wall slip	slip	slip	slip
Lower Wall	Moving Wall (20 0 0)	zeroGradient	kqRWallFunction	omegaWallFunction
Side Wall	Stationary Wall slip	slip	slip	slip
Symmetry Plane	symmetryPlane	symmetryPlane	symmetryPlane	symmetryPlane

Table 2: Boundary conditions for the simulation.

the *snappyHexMesh* application. To be specific, in order to achieve a smooth transition from the external domain to the thin layers above the surface and avoid mesh-dependent results, five regions with increasing levels of refinement were introduced, as shown in Figure 2. The maximum refinement level imposed on the object surface is equal to 8.

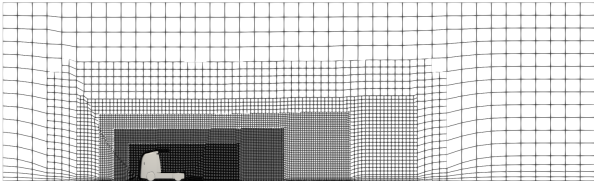


Figure 2: Refinement boxes detail

Moreover, some additional boxes were introduced in the regions where separation occurs, which require a finer grid to capture the physics of the flow. This was done for the front and back of the cabin, the floor, the wheels, the vehicle's tail, and the lateral and upper deflectors. In particular, another refinement region was added for the mirrors, since these components constitute a source of large flow separations. Furthermore, the dimensions and positions of the refinement boxes inside the domain were properly adapted for each configuration.

To reproduce the contact region between the wheels and the lower wall, the latter were positioned in such a way as to slightly intersect the wheel surface [4], thereby creating a tire contact patch that more accurately reproduces the phys-

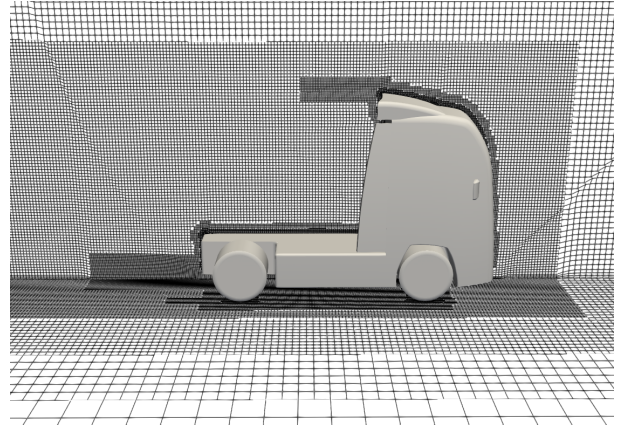


Figure 3: Final mesh

ical fluid phenomena occurring in its vicinity. After all these considerations, the final step is to validate the mesh by performing two fundamental studies: farfield independence and mesh convergence.

3.3. Farfield Independence

In order to verify that the domain size does not affect the results due to the blockage effect related to the presence of the truck, a farfield independence analysis was performed. In particular, this study was carried out by modifying the external domain dimensions without changing the grid around the vehicle.

Therefore, given the baseline case, characterized by $80 \times 16 \times 24$ [5] dimensions referring to the length, width, and height of the domain, respectively, three additional cases were considered:

two of them increased the domain size and one reduced it. Then, for each CFD simulation, the resulting drag coefficient was compared.

Dimensions [m]	C_D [-]	ΔC_D [%]
$60 \times 12 \times 18$	0.4163	3.58
$80 \times 16 \times 24$	0.4060	1.00
$100 \times 22 \times 32$	0.4008	0.29
$120 \times 30 \times 40$	0.4019	–

Table 3: Farfield independence Results

The results of the farfield analysis are reported in Table 3 in terms of the C_D and the percentage difference with respect to the largest domain. It is possible to note that the first grid, which is very small, shows a significant difference compared to the reference domain, while the one selected for the analysis returns a negligible difference; therefore, it allows for further savings in computational cost.

3.4. Grid Convergence

Finally, the grid convergence study can be performed by checking that, as the number of cells increases, the drag coefficient converges to a certain value. In order to perform this analysis, five different meshes have been created, gradually increasing the refinement level of the grid, starting from about 800 thousand cells up to 8.2 million cells.

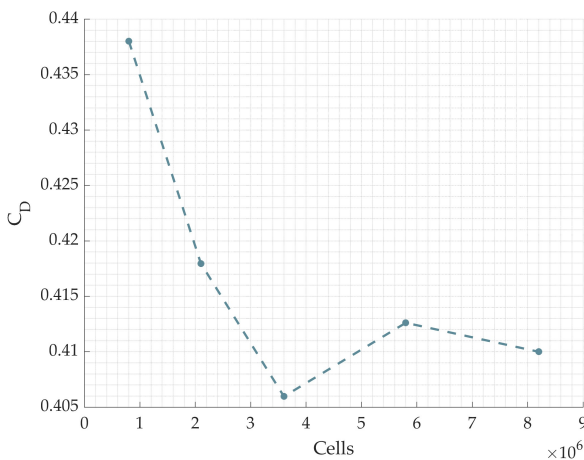


Figure 4: Mesh convergence

Results are shown in Figure 4, where it is possi-

ble to observe that the drag coefficient tends to converge towards a value of approximately 0.41. Looking at this trend, it is reasonable to consider the mesh characterized by 3.6 million cells as a reference for the analysis; this choice allows for a grid refined enough to capture the physics of the flow while controlling the computational cost.

Note that the present analysis was performed considering the geometry with all devices installed, since it represents the most complex configuration. This guarantees that the study of the other cases yields numerically reliable results, even if the number of cells slightly changes.

4. Simulations results

Considering the numerical set-up previously described, it is possible to proceed with the analysis of the problem. In particular, since the main goal of the study is to evaluate how the addition of the cabin roof and side fairings affects the overall drag of the vehicle, the empty³ configuration is considered as the baseline.

Therefore, a CFD simulation of the empty truck was performed first; then, the different devices were installed one at a time to evaluate the impact of each on the vehicle's performance.

Finally, once the results are obtained, the aim is to find aerodynamic solutions to improve the baseline performances and reduce the negative impact of the fairings.

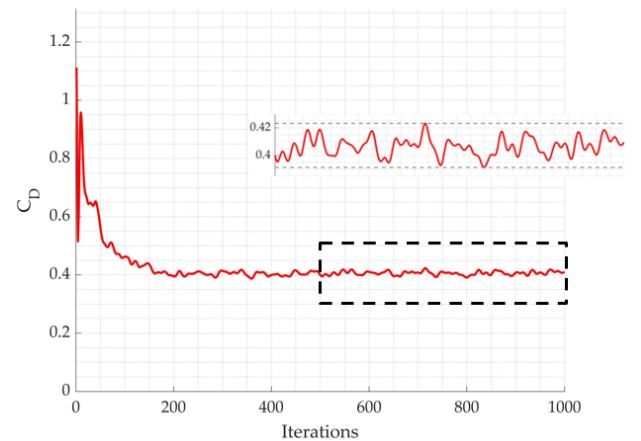


Figure 5: Drag coefficient convergence behavior

This analysis has been performed by examining both numerical results, in terms of the drag co-

³Without any aerodynamic device installed.

efficient, and flow visualizations, in order to understand the physics of the phenomena.

An example of the drag coefficient convergence is reported in Figure 5. In particular, it can be observed that after about 200 iterations, the C_D stabilizes with acceptable oscillations around its reached value, as anticipated in subsection 3.1. In fact, the oscillations arise from the fact that a bluff body generates an intrinsically unsteady flow, which causes the drag not to perfectly converge. For this reason, the C_D is always computed in post-processing as an average of the last 200 iterations of the simulation.

4.1. Preliminary analysis

The four configurations considered are:

- Empty: no aerodynamic devices installed;
- Side fairings⁴;
- Roof fairing;
- Complete: both side and roof fairings;

The results are presented in Table 4.

Configuration	C_D [-]	ΔC_D [%]
Empty	0.3836	—
Side fairings	0.3814	-0.57
Roof fairing	0.4072	6.15
Full	0.4060	5.83

Table 4: Preliminary analysis on C_D

As expected, when all devices are installed, the resulting drag increases by almost 6% with respect to the empty configuration, which represents a significant result for this kind of application. In particular, it seems that the negative effect is mostly related to the roof fairing, while the side fairings actually produce a slightly positive impact.

From Figure 6, it is possible to gain detailed insights into the impact that each component has on the drag force along the truck.

First of all, in all cases, there is a sudden decrease in drag past the front stagnation point due to the curvature of the cabin, which causes the flow to accelerate. This reduction in C_D ends differently whether the roof fairing is present or

not. It is clearly visible that such device has a negative impact on the aerodynamics, since it is responsible for the increase in the wake dimension. The other common features are the two bumps occurring at the wheels, which cause an overall worsening of the aerodynamics.

The other significant differences between the configurations are the effects of side fairings. They, in fact, delay the separation of the flow, which leads to lower drag; otherwise, if they are not present, a sharper increase of C_D occurs.

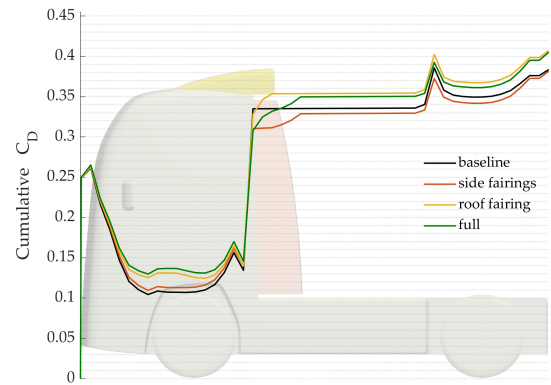


Figure 6: Cumulative drag: preliminary comparison

4.2. Side Fairings

The results reported in subsection 4.1 refer to a configuration characterized by side fairings inclined outward by 5.5° , leading to a slight reduction in terms of drag with respect to the baseline case. A further improvement in the vehicle's performance could be expected when reducing the width of the wake, which is directly affected by the angulation of these devices.

In order to verify this hypothesis, a simulation with a 10° inclination towards the inside was performed, leading to an increase in the drag coefficient by about 0.38%. This result may appear to be misleading, but it can be explained by the presence of the rear part of the tractor, which allows for the connection with the trailer. In fact, by turning the side fairings inward, the flow is conveyed towards this region, where the interaction with the tail of the truck generates strong swirling of the flow, leading to an increase in drag. Such considerations can be appreciated by looking at the Figure 7, which represents a comparison of the streamlines between the two

⁴The configuration considered has an outward angle of 5.5° .

side fairing configurations⁵.

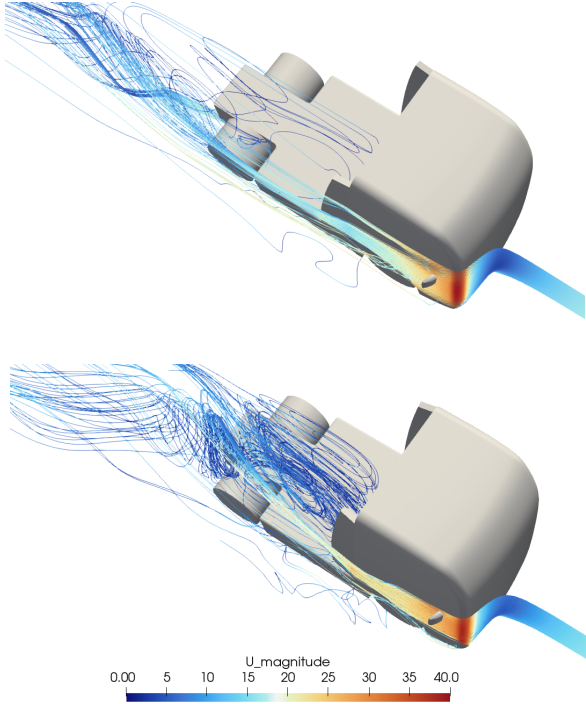


Figure 7: Comparison between streamlines.
Top: 5.5° outward; bottom: 10° inward

The figure clearly shows that, although the wake seems to be more slender, a swirling of the flow arises.

In order to perform a more precise analysis of this phenomenon, a more refined geometry of the tail region of the truck would be required; however, it is not the aim of this analysis to delve deeper into this topic.

Another aspect influencing the results is the development of tip vortices on the fairings, which are generated exclusively when the angle is tilted inward, due to the pressure difference between the two sides of the deflector. The analysis of this phenomenon is further investigated in following sections.

Furthermore, an interesting aspect to consider when analyzing the side fairings is their interaction with the mirrors.

Mirrors

As shown in Figure 9, the wake generated at the height of the mirror interferes with the wall-bounded flow, resulting in the separation of the

⁵The upper image represents the 5.5° case, while the lower one represents the -10° configuration.

flow along the deflector. A possible improvement to this problem could be found by employing mirrorCams instead of traditional mirrors. As shown in Figure 8, the devices were designed employing a *NACA0012* wing profile as the mirror arm to avoid a bluff body behavior of the flow.

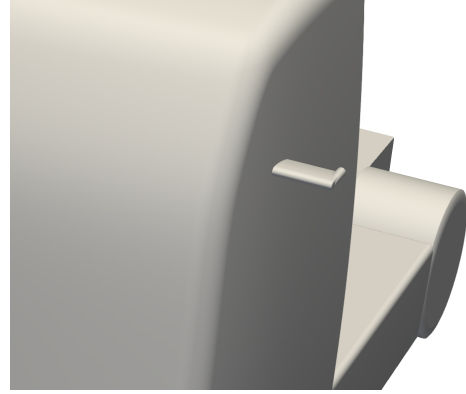


Figure 8: MirrorCams detail

In particular, this configuration, not only improves the quality of the flow going towards the back of the cabin, but allows for a reduction in the device size, resulting in a smaller wake, hence smaller drag. Thanks to this solution, the interaction with the side fairing becomes almost negligible, as shown in Figure 9. Moreover, the use of MirrorCams in the empty configuration allows for a reduction in the total drag, leading to a 1.62% improvement with respect to the reference empty case.

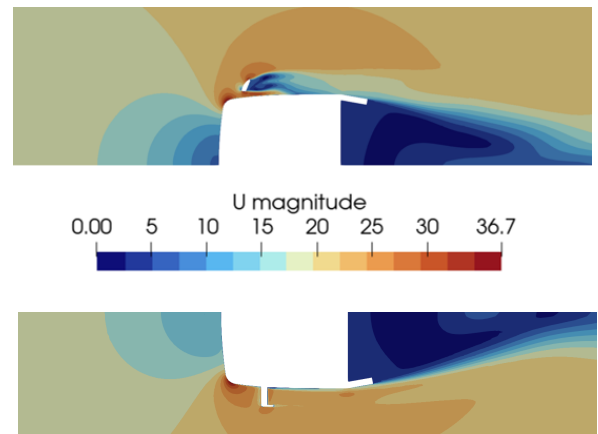


Figure 9: Mirror-wake interaction with side fairings.
Top: standard mirror; bottom: MirrorCam

4.3. Roof fairing

As explained in subsection 4.1, the addition of the roof fairing leads to a significant detriment of

the aerodynamic efficiency of the vehicle. This effect can be related to the fact that this component induces a thickening of the wake, as shown in Figure 10, which increases drag. Therefore, it can be expected that a reduction in the thickness of the wake should provide an improvement with respect to the empty and roof fairing configurations.

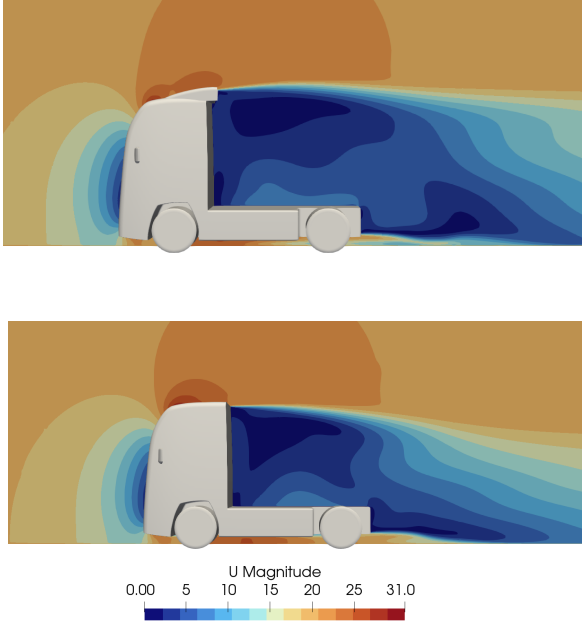


Figure 10: symmetry plane wake difference between baseline and roof fairing configuration

An initial idea was to create an opening at the front of the roof fairing to allow the injection of air inside the wake of the truck, as shown in Figure 11.

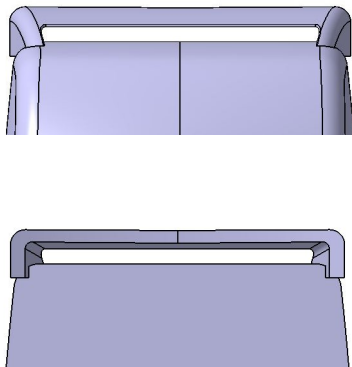


Figure 11: Roof opening geometry

However, this solution did not bring any improvement; it actually led to a 6.99% increase in drag with respect to the baseline, which is

comparable to the standard roof fairing configuration.

The explanation behind this result can be found by comparing the cumulative drag in the two configurations just presented.

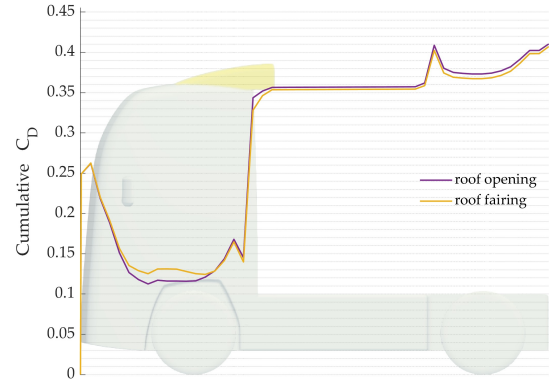


Figure 12: Cumulative drag: standard vs open roof fairing

Figure 12 shows that an opening is actually beneficial for C_D along the cabin; yet the complex geometry at its end is overall detrimental. Further investigations revealed that the fast flow coming out of the opening interacts undesirably with the slower flow in its surroundings, compromising the effectiveness of the solution.

Another possible solution was considered by introducing a simple deflector at the end of the cabin roof, inclined at an angle equal to 10° , as shown in Figure 13. The idea is that the flow, without separating, should follow the walls and then detach, allowing to delay the separation and reduce the dimensions of the wake behind the cabin. However, this resulted in a 3.93% increase in the drag coefficient with respect to the reference configuration, and the reasoning for such behavior can be related to the physics of the flow generated by this device. In particular, when dealing with three-dimensional plates with a certain angle of attack, the pressure difference between the upper side and the lower side generates a recirculation region at the tip due to the motion of the flow from the high pressure region to the low pressure one. As a result, in similar fashion to three-dimensional wings, two large vortices are induced at the edges of the deflector, as shown in Figure 13. These vortices generate regions of low pressure extending along the wake, leading to an increase in the drag coef-

ficient. In particular, the figure shows the total pressure loss coefficient[4], defined as:

$$C_{P_{T,L}} = 1 - C_{P_T} = 1 - \left(\frac{P}{\frac{1}{2}\rho U_\infty^2} + \frac{U^2}{U_\infty^2} \right)$$

which is a measure of the energy loss of the flow due to the presence of the wake.

The solution exploited, however, can be investigated further: in the upper-central part, the wake tends to shrink, meaning that the flow is actually deflected downward, as anticipated.

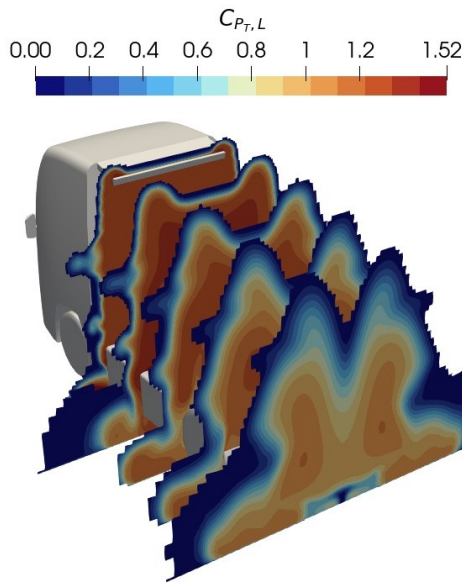


Figure 13: Wake on deflector configuration

From this reasoning, the introduction of side fairings could be expected to limit the development of those vortices, acting in a similar way to winglets at the tips of airplane's wings.

4.4. Side-top fairings interaction

The generation of the two tip vortices due to the presence of the deflector could be limited by introducing the side fairings previously analyzed in subsection 4.2; as already suggested, they should reduce the recirculation of the flow from the upper to the lower side of the deflector. In order to verify this hypothesis, a CFD simulation with the addition of the side fairings inclined at an angle of 5.5° to the roof deflector configuration has been performed.

In particular, the simulation resulted in a 1.4% increase in the drag coefficient with respect to the baseline, representing a significant improve-

ment compared to the configuration without side fairings. As a consequence, this result proves the starting hypothesis that the presence of the side fairings limits the generation of vortices behind the cabin.

To further investigate the interaction between these two devices, a configuration in which the side fairings are inclined at a null angle with respect to the cabin has also been considered; in fact, a reduction in the spacing between the two components leads to further inhibition of the vortices.

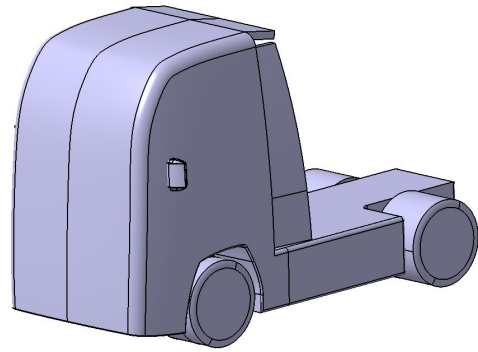


Figure 14: Deflector geometry at 10° with side fairings at 5.5°

The final result shows a -1.92% reduction in C_D ; which is a significant improvement with respect to the previous configuration.

Pressure recovery and flow separation

Another investigation concerns the analysis of the flow past the roof deflector as its angle changes, starting from 0° to 20° , while keeping the side fairings at a 0° angle. From the results shown in Table 5, a singular trend emerges. The best configuration seems to be the one with a straight deflector. This solution employs the same design principle as that found in hatchback cars. Similarly to this application, the truck cabin is characterized by a limited streamwise dimension with a continuous curvature on the roof that generates an adverse pressure gradient. For this reason, if the flow reaches the deflector with a certain non null angle, it continues to decrease its pressure. This ultimately causes the base pressure to reduce, resulting in a larger pressure drag. If, instead, the deflector is kept straight, the flow recovers pressure, and thus a positive outcome on drag is obtained.

Angle [°]	C_D [-]	ΔC_D [%]
0	0.3591	-6.34
5	0.3657	-4.68
10	0.3763	-1.92
15	0.3901	1.68
20	0.4063	5.92

Table 5: C_D for different roof deflector angles

Clearly, the more the deflector is extended, the better is its effect; however, its geometry is constrained by the presence of the trailer.

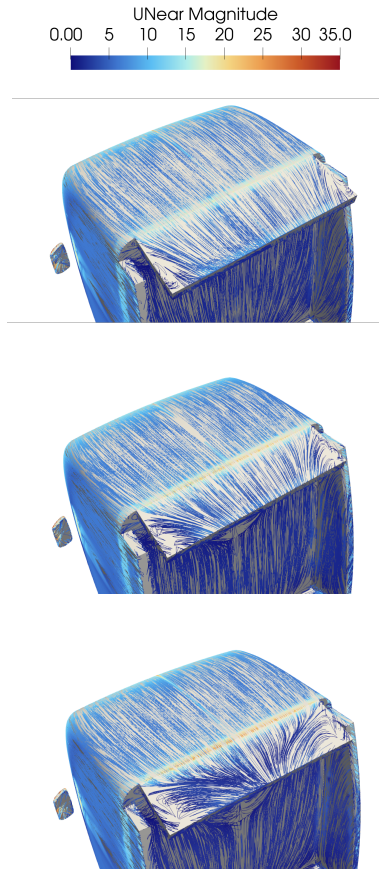


Figure 15: Wall bounded streamlines on the deflector: top 10°; middle 15°; bottom 20°

Note that this reasoning is incorrect if applied to the trailer end itself; in that case, since the flow has run across a straight wall for a long distance, it is able to recover the pressure. In fact, for that application, the classic boat tail solution with fairings deflected towards the center allows for a reduction in the wake size and, as a

consequence, in the drag.

An additional interesting consideration to explain the results coming from Table 5 can be made by looking at the wall bounded streamlines⁶ on the deflector.

Comparing the three cases at 10°, 15° and 20° in Figure 15, it is possible to see that at the tip, there is a small region of separated flow caused by the vortices previously described. This region expands as the angle of attack increases, since the vortices become stronger and stronger. This last reasoning explains why an increase of the deflector angle yields a negative impact on the overall aerodynamics of the truck.

4.5. Boat Tail

As a result of all previous considerations, a final configuration is examined: a boat tail characterized by side and roof fairings at 0°.

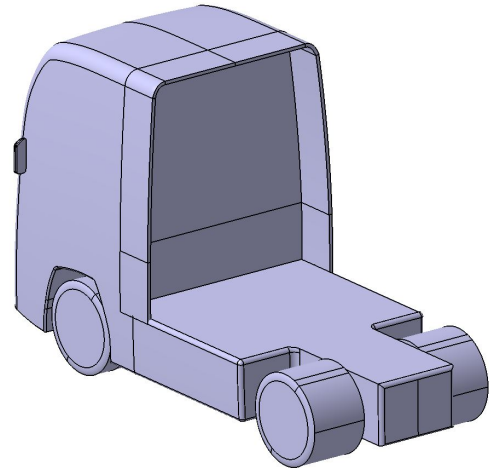


Figure 16: Boat-tail geometry

Although this configuration may not be easily applicable in practice, it is particularly helpful for validating all the previous considerations. To be specific, in this case, the boat tail acts as a simple extension of the cabin, which helps to recover more pressure and delay the separation, hence obtaining a higher base pressure, rather than trying to reduce the dimension of the wake. This results in a 7.18% drag reduction compared to the empty configuration.

From Figure 17, which shows the cumulative drag, it is possible to observe that the two curves

⁶Streamlines developing in close proximity to the wall (1 mm).

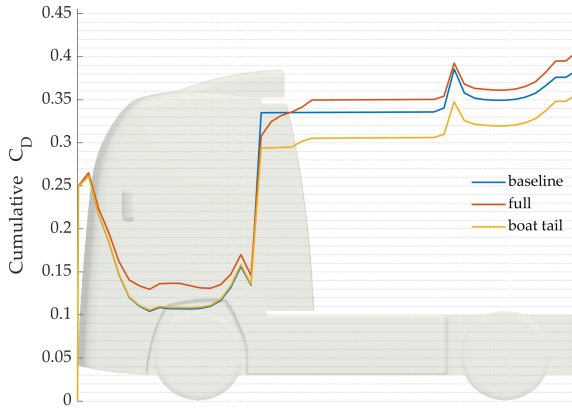


Figure 17: Cumulative drag: boat tail analysis

for the empty and boat tail configurations are characterized by a lower and almost identical drag coefficient along the cabin. At the end of the cabin the separation occurs, creating a large wake; however, it is possible to note that this phenomenon occurs in a different way, depending on the geometry. Looking at the empty configuration, there is a clear peak corresponding to the end of the cabin; whereas for the other two cases, the drag coefficient increases further due to the presence of the fairings, which delay the separation. However, between these two, there is a significant difference; the pressure recovery obtained with the boat tail configuration allows for a lower increment compared to the configuration with all fairings installed.

5. Conclusion

To summarize, this work has investigated the effect of drag-reduction devices installed on heavy-duty trucks on their aerodynamic efficiency whenever the trailer is not present. The analysis was first conducted by examining the combined influence of all devices on the vehicle's drag coefficient, and then by analyzing the contribution of each component individually.

In particular, the roof fairing has been determined to be the most critical element, as it generates a substantial increase in drag. Therefore, several configurations have been analyzed in order to understand the physics of the wake that develops behind the cabin. This investigation ultimately led to the identification of an effective solution to improve the aerodynamic behavior of the vehicle: a straight boat tail.

The analysis carried out, in conclusion, shows that, to obtain an aerodynamic improvement on a truck tractor, it is more effective to focus on recovering the base pressure rather than finding solutions to reduce the wake.

Configuration	C_D [-]	ΔC_D [%]
Empty	0.3836	–
Full	0.4060	5.83
Boat tail	0.3561	-7.18

Table 6: Final C_D comparison

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